August 2, 2026

Alberta Machine Intelligence Institute (Amii)

Dear David Staszak, Amor Provins, and Hiring Team,

I am writing to apply for the Machine Learning Resident - Client: Medtronic (12 month term) position with Amii. I am a Master of Science in Computer Science graduate from the University of Calgary with hands-on experience in Python, PyTorch, deep learning, model evaluation, data-processing pipelines, and research communication. The opportunity to apply machine learning to pressure injury prevention is especially compelling to me because it combines rigorous predictive modeling with a clear clinical and human impact.

My graduate research has focused on building reliable machine learning and data-management workflows for large, complex datasets. In my DEM super-resolution work, I trained and evaluated PyTorch ResNet models, built data collection and preprocessing pipelines with Streamlit, Rasterio, and Google Earth Engine, and assessed model behavior using RMSE, slope, TRI, TPI, and wavelet-based metrics. This work required careful dataset construction, baseline evaluation, model experimentation, validation design, and clear communication of tradeoffs and results.

I also developed a DGGS-based framework for real-time multiresolution management of spatiotemporal Earth observation data through my BigGeo/Mitacs research collaboration. Although the domain was geospatial rather than healthcare, the project strengthened skills that align closely with this residency: ingesting and organizing data from multiple sources, designing reproducible pipelines, validating outputs, optimizing performance, and translating research ideas into working software. My first-author publication from this work is the professional accomplishment I am most proud of, because it represents several years of independent research, engineering, experimentation, writing, and collaboration brought to a peer-reviewed result.

Beyond research, I have experience explaining technical concepts to diverse audiences. As a teaching assistant in deep learning, big data, data-at-scale, and database courses, I helped students reason through models, pipelines, SQL, cloud workflows, and project implementation. I enjoy collaborative technical work and would be comfortable preparing reports, presenting findings, and working iteratively with domain experts to understand features, evaluation goals, and deployment expectations.

I would bring Amii a strong foundation in PyTorch model development, Python engineering, data analysis, reproducible experimentation, and scientific communication, along with genuine interest in learning the healthcare domain responsibly. I would welcome the opportunity to contribute to a predictive tool that helps care teams identify patients at higher risk and intervene earlier.

Thank you for considering my application. I look forward to the opportunity to discuss how my background can support Amii, Medtronic, and this machine learning residency.

Sincerely,
Amir Mirzai Golpayegani
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